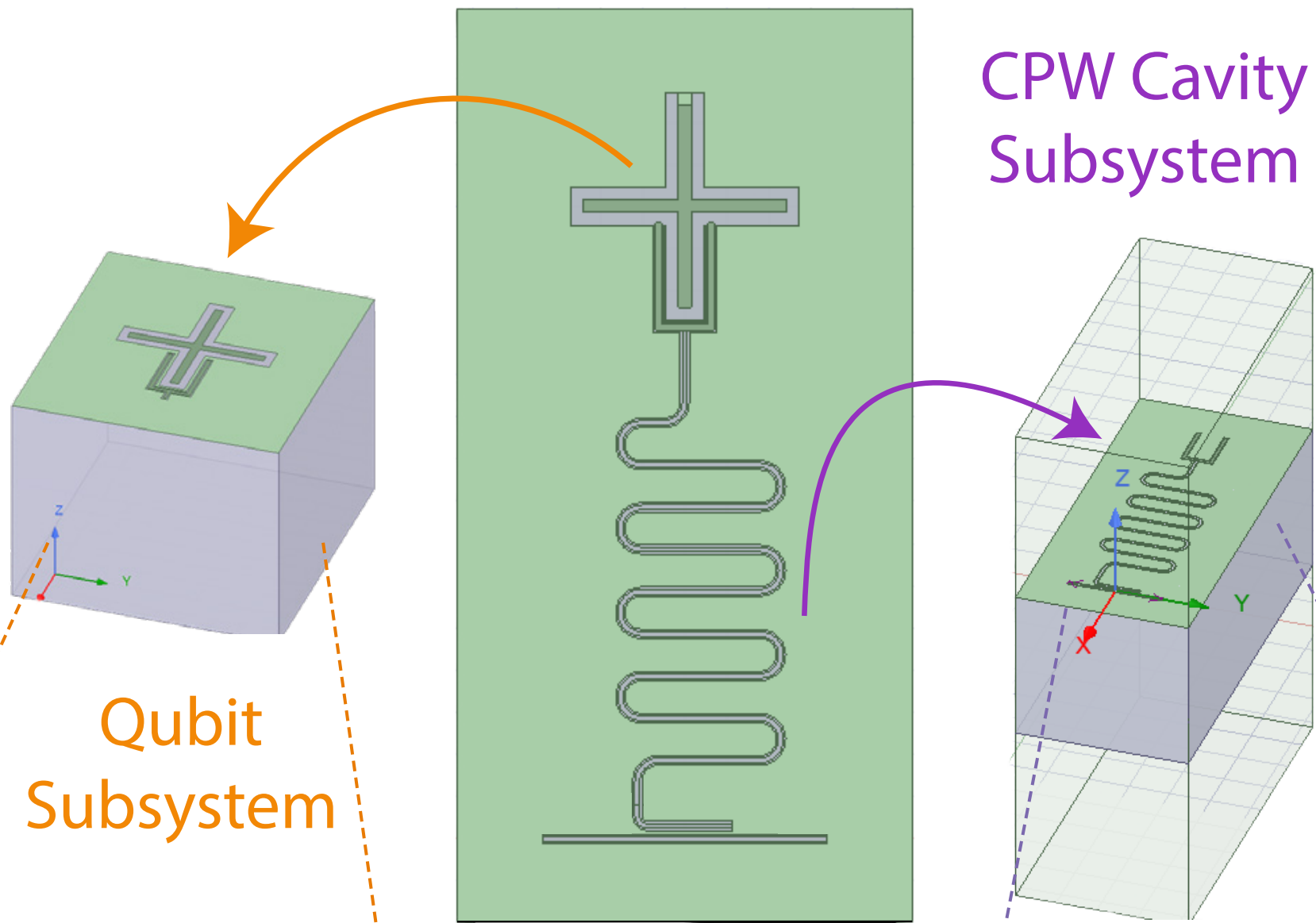




CPW Cavity
Subsystem

Qubit
Subsystem

α (anharmonicity)
 $\approx -f_C$ from Q3D

f_q (qubit frequency)
 $\approx \sqrt{8 f_J f_C} - f_C$ from Q3D

f_r (resonator freq.)
HFSS eigenmode solver

κ (resonator linewidth)
HFSS via 50 Ω port loss